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3D brain tumor segmentation in MRI images using hierarchical adaptive pruning of non-tumor regions

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ABSTRACT

Background: The detection of brain tumors in MRI images has significantly improved with the advent of deep learning methods. However, these approaches often suffer from high complexity, computational cost, and the need for extensive annotated training data, making them less practical for real-time and patient-centered diagnostic systems. To address these challenges, this study introduces a perceptually inspired, algorithmic method that mimics the diagnostic behavior of physicians, offering a lightweight and interpretable alternative for brain tumor segmentation.

Method: We propose a novel adaptive hierarchical pruning algorithm for 3D MRI brain images that iteratively removes low-intensity, non-tumor voxels based on the statistical distribution of intensities. The tumor region is identified through the comparison of the remaining pixel intensity values statistics. The pruning automatically stops when the mean and median of the remaining voxels converge, leaving the candidate tumor region.

Results: The proposed algorithm was evaluated on all patients of the BraTS2019 and BraTS2023 datasets, achieving segmentation accuracies of 99.1 % and 99.13 %, respectively. It demonstrated high sensitivity and specificity compared to several deep learning methods, showing robust performance across diverse patient scans.

Conclusions: This study demonstrates that a simple, perceptually driven segmentation algorithm can match or outperform complex deep learning models, particularly in clinical settings where lightweight, transparent, and efficient tools are essential.

1. Introduction

Brain tumors are among the most aggressive and life-threatening cancers, making accurate segmentation in magnetic resonance imaging (MRI) critical for diagnosis, treatment planning, and prognosis. Manual segmentation is time-consuming and relies heavily on radiologist expertise, prompting the development of automated methods.

Deep learning, especially convolutional neural networks (CNNs), has achieved promising results in tumor segmentation. Advanced models such as CNN-BiLSTM with evolutionary optimization [1], wavelet-enhanced CNNs [2], and GAN-based survival prediction [3] have been proposed. Platforms like NeuroVision [4] integrate these models into clinical workflows.

While most recent studies in medical image segmentation have shifted toward deep learning-based approaches, such methods are highly dependent on large-scale annotated datasets, complex network architectures, and extensive validation processes. In contrast, our proposed method is a fully analytical, image- and tumor-characteristic-based approach that requires no training data. It can be directly implemented on 100 % of the dataset and consistently produces reliable segmentation results. This inherently eliminates the dependency on training datasets and the risk of overfitting. The critical knowledge gap lies in the fact that traditional analytical methods based on the intrinsic properties of medical images have been largely abandoned in favor of data-driven imitation approaches. Our study addresses this gap by demonstrating that leveraging the inherent nature of medical images can provide a simpler, more cost-effective, and equally (or more) effective solution for 3D brain tumor segmentation. This study introduces a novel 3D brain tumor segmentation method using hierarchical adaptive pruning based on statistical intensity analysis to iteratively exclude non-tumor areas, isolating the tumor. Unlike previous methods, this approach identifies the tumor region by only comparing statistics (mean and median) of pixel intensity values remaining after the hierarchical pruning process, which can be considered a semi-supervised learning process. The method is training-free, computationally efficient, and suitable for resource-constrained

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diagnostic settings.

The contribution of this work can be summarized as follows:

- A novel hierarchical pruning algorithm inspired by physician perception.
- A novel tumor segmentation mechanism based on the convergence/comparison of mean and median voxel intensities after pruning, distinguishing it as a somewhat semi-supervised learning approach.
- Evaluation on BraTS2019 and BraTS2023 datasets with comparisons to deep learning models.
- Comprehensive graphical and statistical analysis of segmentation performance.

The paper is organized as follows: Section 2 reviews related work; Section 3 describes the proposed methodology; Section 4 presents the experimental results; Section 5 discusses the findings; Section 6 addresses limitations, real-world dataset validation, and computational complexity; Section 7 provides a comparative evaluation with deep learning models; Section 8 discusses practical implications; Section 9 concludes the study; and Section 10 provides the author declaration on AI usage.

2. Review of related works

The most common and well-known method for early diagnosis of brain cancer is through the use of magnetic resonance imaging [5]. In some cases, further imaging tests such as PET[1] and CT are used in combination with MRI. Abnormalities in light intensity in a particular area of the brain can indicate a tumor [6,7]. By identifying the problematic area in the images, neural tests can be used to confirm or make a more reliable diagnosis. Depending on the area of the tumor diagnosis, the relevant specialist checks the patient's hearing, vision, coordination, balance, strength, and reflexes. In some cases, the specialist may use an invasive biopsy to provide a final confirmation. This method involves collecting a sample of problematic or abnormal tissue that is examined microscopically [8,9].

MRI imaging can be a valuable tool for diagnosing medical conditions and creating personalized treatment plans. During an MRI scan, a contrast agent or dye may be injected into the bloodstream. This allows medical professionals to gain a better understanding of the patient's condition and use different types of MRI scans (such as perfusion MRI, functional MRI, and magnetic resonance spectroscopy) to formulate a customized treatment plan [6,7].

An accurate and timely diagnosis of a disease is crucial for a patient's survival. However, identifying malignancies can be a challenge for physicians due to the large amount of image data and the uncertain morphological structure of tumor areas in these images [10]. To help diagnose cancer more accurately, physicians have started using CAD tools [11–14]. Brain MRI images can be processed by computers and displayed to physicians, which increases the accuracy of image analysis and the diagnosis of brain tumors [15]. Recently, researchers in the field of artificial intelligence have developed an approach that simulates how experts analyze images. This approach can reduce detection time and memory requirements for image storage [16,17]. If the accuracy and generalization capability of this image analysis system is at the level of an expert doctor, its use will reduce the time and memory volume and ensure the absence of human error in diagnosis. Another benefit of using a diagnostic system is that it can extract hidden knowledge from big data. For example, a radiologist can collect more details about normal and tumor tissues using such a system [18]. Data mining algorithms can also analyze the imaging and/or non-imaging data of the patient simultaneously. In the past few decades, this technology has significantly impacted the early detection of cancer and its timely treatment [19].

Image segmentation is a widely used method for analyzing images, which complements auxiliary diagnostic systems and transforms them into diagnostic systems. Medical image segmentation algorithms use brightness, color, or texture features to divide image regions into anatomically meaningful regions [20–23]. This creates diagnostic attractions for physicians and also serves as a form of image compression. There are various methods used for the segmentation of medical images, such as multi-level thresholding algorithms, clustering (KNN,[2] K-means, FCM,[3] Mean Shift, GMM,[4] and hierarchical clustering), and classification (Simple Bayes, ANN,[5] SVM,[6] random forest, decision tree, vector quantization learning, DBSCAN,[7] and deep learning-based methods) [7].

The diagnosis and classification of brain tumors in magnetic resonance images are based on the difference in the visual characteristics of the tumor area with other areas of the image and the difference in the visual characteristics of the tumors (such as stroke, benign, malignant) with each other. Various approaches to express and recognize this difference in visual characteristics have been presented. These approaches are outlined in Table 1.

Algorithms that use thresholding to detect and segment tumors assume that the brightness intensity values of the tumor region are different from those of other areas in the image. On the other hand, clustering algorithms are used for image classification by locating similar points based on certain features [24]. In the case of tumor segmentation, classification algorithms are used in the first step after extracting the distinguishing features of tumor and non-tumor regions in MRI images.

A simple approach for tumor segmentation involves using the median filter and Otsu thresholding [25]. However, due to the nature of MRI images and the limitations of Otsu's thresholding method, some non-tumor areas with high brightness intensity may be mistakenly identified as tumors. This issue can be resolved by performing morphological post-processing to obtain the exact tumor region. Surbhi et al. proposed an alternative approach that involves using a diffusion filter and Otsu thresholding for feature extraction. They then trained a neural network to detect the tumor region, eliminating the need for morphological post-processing [26]. In some sources, the multilevel thresholding method has been used to segment the MRI image into different tissues, including tumors. However, morphological post-processing is still required to remove unwanted areas [27].

Different clustering algorithms can be utilized in the development of the unsupervised thresholding method to differentiate between the tumor and other areas of the image. KNN algorithm, K-means, and its fuzzy version (FCM) have been used for this purpose. The KNN clustering algorithm is useful for classification because it allows for easily adding new training data. This algorithm does not summarize the relative position of the training data in the feature space, does not extract the decision boundary, and is lazy in giving depth to learning. During the training phase, this algorithm maintains only the positions and labels of the points with full laziness and high speed. Therefore, it is reasonable to have more calculations in the estimation phase, making this method very fast and flexible in learning, but slow in estimation. This algorithm has been used in diagnosing brain tumors and differentiating stroke, benign tumors, and malignant tumors [28–30].

The K-means algorithm assumes that similar points in the feature space are close to each other. However, unlike the KNN algorithm, K-means summarizes all the training information as the K positions of the cluster center in the feature space. This algorithm and its fuzzy version

[1] Positron Emission Tomography.

[2] K-Nearest Neighbors.

[3] Fuzzy C-Means.

[4] Gaussian Mixture Model.

[5] Artificial Neural Network.

[6] Support Vector Machines.

[7] Density-Based Spatial Clustering of Applications With Noise.

Table 1
Different approaches to 3D tumor detection in MRI.

	thresholding	Classification	clustering
Hypothesis	The brightness of the tumor area is different from other areas of the image.	It is possible to extract distinguishing features of tumor and non-tumor areas in MRI images.	Similar points in terms of some features can be found together in the space of these features.
Method(s)	Preprocessing Apply Otsu bi/multilevel thresholding Postprocessing	feature extraction (conceptual algorithms/CNNs) and classification (conceptual algorithms/CNNs)	KNN, K-means, FCM, Mean Shift, Hierarchy, GMM

Combination of classification and clustering
K-means clustering and kernel-based support vector machine (KSVM) [44] Combination of CNN and SVM [47] small kernels of two-path convolutional neural networks and random forests [46]

Combination of thresholding and classification
Using a diffusion filter and Otsu thresholding for feature extraction and neural network training for tumor area detection [26]

(FCM) are mostly used for clustering the brightness intensity values of medical images, but can also be used for segmentation and classification [31,32]. It works much faster than KNN in estimating the data class or category, and the closest cluster center to the input data determines its class. However, compared to the KNN algorithm, adding new training data in the K-means algorithm is not as straightforward [28].

The moving average clustering algorithm, also known as the mode search algorithm, is a powerful technique that doesn't require any parameters. It is useful in various unsupervised learning scenarios [13]. This algorithm moves each data point to the average of the nearby points [33]. In brain tumor detection from noisy images, this method was combined with gray-level co-occurrence matrix information for automatic detection [34]. The results of using FCM clustering on the moving average output were also satisfactory for brain tumor segmentation [35]. In a short time, the use of K-means and improved FCM was also successful in detecting human brain tumors [31]. For brain tumor diagnosis, hierarchical clustering has been used to merge similar clusters in later stages [36].

A new Association Assignment Hierarchical Clustering approach was proposed for early detection of brain cancer and achieved 100 % accuracy while being computationally efficient [37].

An algorithm for brain tumor segmentation was developed using GMM and the back-jump Markov chain Monte Carlo algorithm. This algorithm is fast and efficient [38]. Another approach was taken to cover the limitations related to the Gaussian mixture model and the generalized Gaussian mixture model. This approach used adaptive neonormal distributions in the form of a mixture model. This model was able to cover the symmetric and asymmetric patterns of an MRI image for brain tumor segmentation—the adaptive neonormal distribution presented better segmentation results [39].

The researchers used IMS[8] to analyze samples of brain tumors. They tested two methods of analysis - IMS region of interest and new hierarchical clustering, with the hope of identifying biological markers in brain cancer samples. These methods show promise in finding useful information for brain cancer research [40].

Supervised learning techniques have been utilized in medical research to detect tumor regions in MRI images. In Ref. [41], researchers used discrete wavelet and directional gradient histograms to extract features, and then employed classifiers such as random forests, decision trees, and support vector machines to classify the image into normal and abnormal. Brain tumor identification and segmentation have also been performed using random forest, as described in Ref. [42]. Finally, a comparison of brain tumor diagnosis and segmentation has been made between decision trees, support vector machines, and random forests [41].

Various techniques have been employed to accurately detect and locate tumor pixels. One such approach involves combining unsupervised learning and supervised learning methods. In this method, supervised learning is simplified by using an initial clustering. The image is first classified into normal and abnormal using a support vector machine and adaptive neural fuzzy inference, and then the tumor is segmented using fuzzy cluster centers and related optimizations [43]. Another method is to use a combination of K-means clustering and kernel-based support vector machine (KSVM) for automatic brain tumor segmentation and classification [44]. Although SVM can work well even with semi-structured and unstructured data using a suitable kernel function, finding a good kernel function is not an easy task.

In one study, local histograms were used for feature extraction, and both support vector machines and random forests were employed for medical tumor detection [45]. Another study employed convolutional neural networks for feature extraction, using small kernels of two-path convolutional neural networks and random forests for classification [46]. A combination of CNN[9] and SVM was used for feature extraction and classification in yet another study [47].

Convolutional neural networks, and deep neural networks in general, have demonstrated high accuracy in identifying brain tumors in MRI images. The model can learn numerous patterns by adjusting the many weights and biases in its convolutional layers. This process is accomplished by training the network with a large dataset, which plays a crucial role in achieving optimal results. These neural networks are the

[8] Imaging Mass Spectrometry.

[9] Convolutional Neural Networks.

best option for modeling data sequences and the complex dynamics of human behavior. However, they are prone to overfitting and require a large amount of training data, high memory, bandwidth, and time for training. Additionally, analyzing images from different imaging devices than that used in the training process may present challenges [48].

Previous studies have demonstrated satisfactory accuracy in brain tumor segmentation using CNN. However, these studies have failed to address the issues raised by the size and complexity of network structures in medical image analysis. This is particularly problematic for portable diagnostic and therapeutic devices that require a simple structure. Some of the issues that can be highlighted include:

Automatic recognition systems for medical images can take the form of online applications that can work with these images. These applications can be web-based or mobile. By designing a simple algorithm, one can obtain real-time image analysis results. The new generation of healthcare systems is being developed to be more patient-centered, and therefore, low-power and simple algorithms must be devised to meet real-time needs.

Algorithmic techniques for segmenting brain tumors, such as perceptron multilayer neural networks, have a simpler structure than convolutional neural networks. However, CNNs perform better in medical image analysis. There are two solutions to make the structure suitable for implementation on embedded devices. First, simplifying complex but efficient structures such as CNNs, and second, increasing the efficiency of simple algorithmic structures.

The accuracy of tumor segmentation in algorithm-based methods will increase with the evolution of imaging systems and the improvement of image contrast, unlike the training-based methods.

One of the challenges of deep learning is its dependence on rich, high-quality datasets. Creating a 3D image dataset and extracting tumor volumes is costly, time-consuming, and requires specialized expertise. Another approach that employs deep learning utilizes generative adversarial networks (GANs) to augment training datasets. A common GAN-based strategy is to generate entire 3D image volumes instead of focusing on specific regions of interest (ROIs). Recently, a new method has been introduced to address unbalanced datasets, particularly for rare types of brain tumors. This innovative approach generates data specifically for the ROI rather than creating the entire brain slices, which enhances the diagnostic process by concentrating on the most relevant areas. However, the authors acknowledge some limitations in their method, particularly concerning the broader challenges of deep learning and the evaluation of its effectiveness [49,50].

Researchers anticipated that the new deep networks designed for segmenting landscape images would be highly effective in tumor segmentation. However, they often found themselves complicating the structure of these networks to achieve the desired efficiency. A common strategy involves incorporating attention mechanisms. By adding an attention convolutional block to the deep network, important features are emphasized while irrelevant features are suppressed [51–53,57,58].

This approach aims to ensure that the network focuses on the tumor region and ignores other areas of the image. However, it remains questionable whether this perceptual goal can be effectively achieved through training alone or whether it will require other structural modifications. Additionally, the effectiveness of the deep network layers in selecting various features from a correlated mass—and then choosing to focus on some while suppressing others—raises important questions regarding optimal performance.

Deep networks encounter significant challenges when it comes to utilizing 3D information related to tumors. Instead of leveraging 3D information effectively, they often resort to using 3D data, which introduces numerous issues and complexities. Additionally, researchers sometimes rely on images from various MRI modalities. It's important to recognize that specialist physicians intentionally choose each modality for specific purposes, integrating the information they gather from each source. Therefore, focusing on data integration rather than information integration may limit the effectiveness of these methods in collaborating with experts [51–61].

3. The proposed method

The main steps of the proposed method are shown in the block diagram of Fig. 1. After a 3D preprocessing step on brain MRI images, the main steps of the algorithm include hierarchical pruning of MRI images, selection of the best-pruned image, and finally, tumor regions are extracted from the best-pruned image.

3.1. Dataset

This study used the Brain Tumor Segmentation (BraTS) public collections to evaluate our method. Specifically, we used the BraTS2019 cohort (335 cases: 76 low-grade glioma (LGG) and 259 high-grade glioma (HGG)) and the BraTS2023 cohort (1250 glioma cases), resulting in a combined total of 1585 cases which employed to verify the high efficiency of the proposed method and its insensitivity to variations in dataset version and therefore imaging device brands and generation [57]. Since our method is training-free, the entire dataset was used for evaluation rather than being split into training, validation, and test subsets.

The BraTS dataset is a crucial resource for research in brain tumor segmentation and has seen numerous enhancements since its creation. These improvements include an increase in both the quantity and quality of data, an expansion to cover more tumor types, and the involvement of various data collection centers. The images in the dataset were obtained using different MRI protocols, such as T1-weighted, T1-weighted contrast-enhanced, T2-weighted, and FLAIR. Each MRI image has been meticulously segmented by neurologists, who identified various tumor regions, including edema, necrosis, and active tumor tissue [57,61].

The patients whose MRI scans were used to create the dataset gave their informed agreement to have their medical information used for research. The data were preprocessed to standardize the size, intensity values, and other imaging characteristics. The MRI scans were taken utilizing various MRI equipment and procedures [41].

The BRATS database uses MRI scans and mainly focuses on 3 major tasks: segmentation of intrinsically heterogeneous brain tumors (Gliomas) in shape, appearance, and histology, overall patient survival prediction, and algorithmic uncertainty evaluation in tumor segmentation.

The inclusion of T1CE MRI images in the BraTS dataset makes the study of enhancing components of BTs possible. This information is crucial for making diagnoses, formulating treatment plans, and tracking the course of the disease.

3.2. Preprocessing

Preprocessing of MRI images is based on two assumptions. The first assumption is that the intensity values of MRI images contain noise, which should be eliminated based on the continuity of the intensity values in all dimensions, including each slice image and also along the slice image axis. The second assumption is that the tumor has a spatial correlation in all three dimensions of the brain MRI image. Image preprocessing is carried out in a way that is consistent with these two assumptions. The simplest method for removing noise and enhancing the spatial correlation of intensity values is to apply 3D Gaussian smoothing of MRI images.

A 3D Gaussian filter with three standard deviations of $7 \times 7 \times 5$ was applied to only suppress high-frequency noise. In this study, we deliberately adopted a minimal preprocessing pipeline to preserve the diagnostic characteristics of the raw MRI scans and to evaluate the hierarchical adaptive pruning framework independently. The rationale for this choice is to ensure that the segmentation results reflect the effect of the pruning strategy itself, rather than being driven by extensive

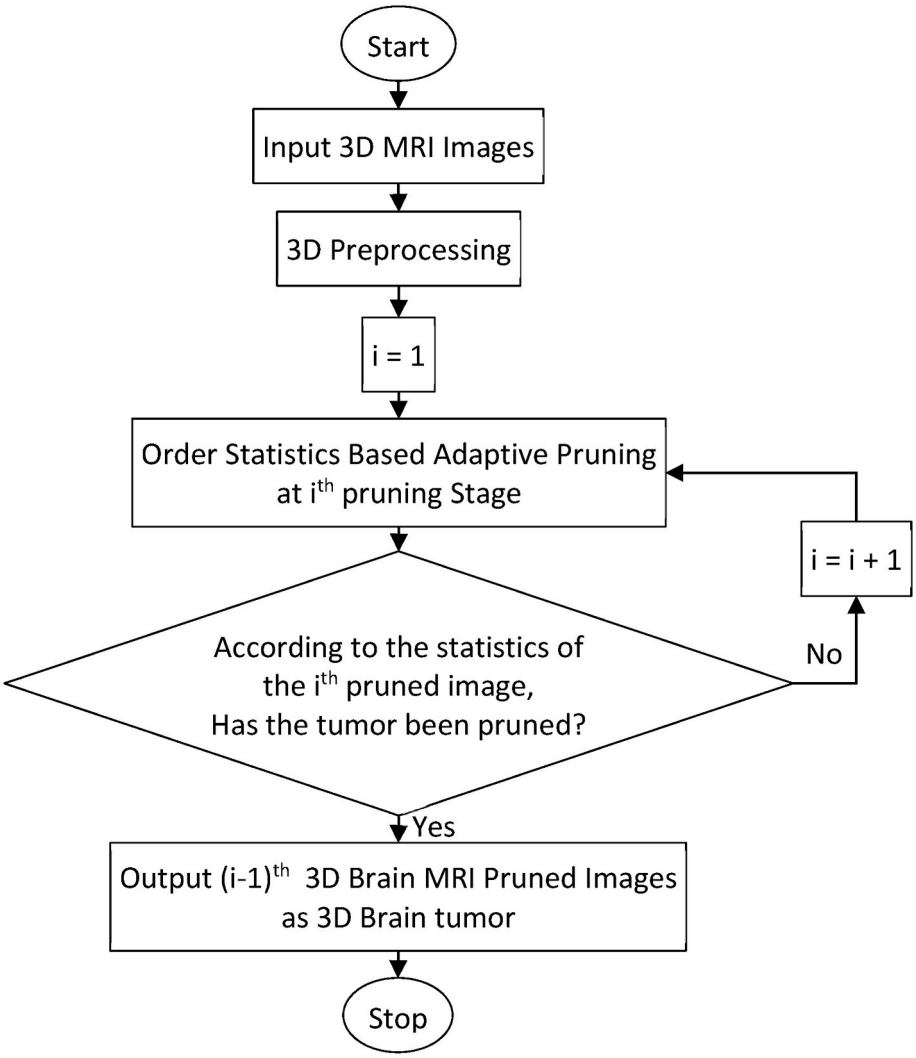


Fig. 1. Block diagram of different stages of the proposed method for brain tumor extraction.

preprocessing. This is also consistent with radiologists' diagnostic practice, where tumor segmentation is typically based on the direct intensity distribution of the images input to the visual system and its Gaussian smoothing in the retina. Due to the small impact of noise smoothing on the values of the evaluation criteria, more effective preprocessing methods, such as Bias-field correction and z-score normalization, have also been used as effective preprocessing methods in various sources, but these preprocessing methods have not been deliberately used in this paper.

3.3. Hierarchical pruning of images

The algorithm involves several key steps, including hierarchical pruning of MRI images, selecting the best-pruned image, and extracting tumor regions. These steps are based on three assumptions. Firstly, the intensity values should be highly uniform in the tumor area, meaning that the mean and median values of the intensity values in the tumor area are almost the same. Secondly, the intensity values in the darker areas of the image are not necessary to locate the tumor area. Thirdly, the intensity values in these darker areas are not uniform, and the mean and median values of the intensity values of these parts are not necessarily equal.

In the process of pruning image intensity values, the first step is to remove all pixels with zero intensity values. After that, the pruning is carried out based on the mean of the unpruned intensity values. This means that in each stage, intensity values that are lower than the mean of the remaining intensities in that stage are removed. From the third step onward, the closeness of the median to the mean is checked to fulfill the stopping condition.

In the first step of pruning, pixels with zero intensity are removed. However, many pixels with intensity values below their mean value have remained. This causes the mean value to be higher than the median value. In the second step, all of these large numbers of low-intensity pixels are removed. This leads to an increase in the median value of the remaining pixels, making it usually higher than their mean value. As we move on to the third step, only high-intensity value pixels are left. Therefore, their mean value is higher than their median value. With each subsequent pruning, the median value gets closer to the mean value.

In the final stages of pruning, due to the uniformity of intensity values in the tumor area, the median of the intensity values is very close to their mean. The relative relationship between the mean and median values of the residual pixel intensities in different stages of image pruning for some selected images from the data set is shown in Fig. 2.

Let $r_k, for\ k = 0, 1, \cdots, L-1$ denote the intensities of an L-level digital 3D MRI image, $f(x,y,z)$. The ith the pruning stage normalized histogram of $f(x,y,z)$ is defined as

$$p(r_k) = \frac{n_k}{M \times N \times S}, k = 0, 1, \cdots, L-1 \quad (1)$$

Where n_k is the number of pixels in $f(x,y,z)$ with intensity r_k, M, N, and S

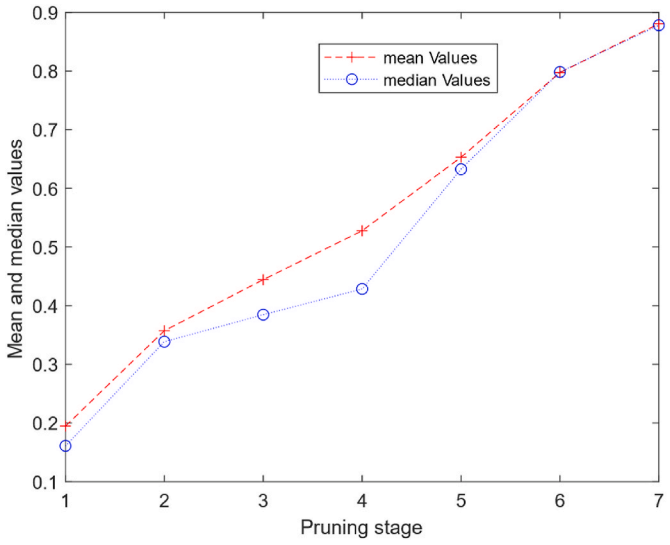


Fig. 2. Mean and median values of residual intensity values in different stages of image pruning.

are the number of image rows, columns, and slices, respectively.

The sum of $p(r_k)$ for all values of k is always 1. The components of $p(r_k)$ histograms are estimates of the probabilities of intensity levels occurring in an image. They are simple to compute and suitable for fast hardware implementations, making histogram-based techniques a popular tool for real-time image processing [62].

All processing is now one-dimensional and conducted hierarchically on the image histogram. At each stage, pruning is applied to the intensity values that remain after the previous stage's pruning.

$$\mu^1 = \sum_{r_k} r_k \times p(r_k) \tag{2}$$

Starting from the second step, the threshold value is determined by the average intensity values of the unpruned pixels from the previous step.

$$\mu^i = \sum_{r_k > th^{i-1}} r_k \times p(r_k) \tag{3}$$

The pruning operation continues until the tumor pixels are pruned according to the condition outlined in the method. Once this condition is met, the tumor structure is obtained by applying the threshold value calculated during the previous step of pruning. The stopping condition is based on the minimal difference between the median value of the remaining brightness values in the ith step and their average value, which is determined by the following equation.

$$th^i = \frac{\mu^i - m^i}{\mu^i} \; ; F(m^i) = \sum_{l=1}^{i} \frac{1}{2^l} \tag{4}$$

In the equation above, $F(r_j)$ represents the probability distribution function of image intensity levels, calculated using the following equation. Thus, m^i denotes the median value of the unpruned intensity values at the ith stage.

$$F(r_j) = \sum_{k \le j} p(r_k), j = 0, 1, \cdots, L-1 \tag{5}$$

3.4. Selecting the best-pruned image

As mentioned in the previous section, in the final stages of pruning, the median and average values of the residual illuminance are almost equal. This equality is due to the greater values of tumor brightness intensity and their relative uniformity. Returning to the previous steps of pruning and where the median becomes less than the mean, the pruned image is considered the tumor region. To see that this choice leads to the best result in locating the tumor region, the regions extracted by this adaptive process are compared with the base regions for the tumor in Fig. 3. As discussed earlier, in the final stages of pruning, the residual illuminance's median and average values become almost equal. This is due to the higher values of tumor brightness intensity and their uniformity. When we go back to the previous pruning steps, and the median becomes less than the mean, the pruned image is considered the tumor region. To ensure that this choice provides the best outcome in locating the tumor region, the regions extracted by this adaptive process are compared with the base regions for the tumor in Fig. 3. As discussed earlier, in the final stages of pruning, the residual illuminance's median and average values become almost equal. This is due to the higher values of tumor brightness intensity and their uniformity. When we go back to the previous pruning steps, and the median becomes less than the mean, the pruned image is considered the tumor region. To ensure that this choice provides the best outcome in locating the tumor region, the regions extracted by this adaptive process are compared with the base regions for the tumor in Fig. 3.

The pseudocode for the Order Statistics-Based Adaptive Pruning Algorithm is as follows. This algorithm aims to accurately segment the brain tumor region via an iterative pruning process. It takes 3D FLAIR MRI scans from the BRATS dataset as input and produces a pruned region that represents the tumor as output.

Algorithm. Order Statistics-Based Adaptive Pruning for Brain Tumor Extraction

```
Input: 3D FLAIR MRI scans from the BRATS dataset.
Output: 3D binary mask of the pruned tumor region.
1. For each subject in the dataset:
2.   Load the FLAIR MRI scan.
3.   Perform preprocessing by applying 3D Gaussian smoothing to the scan.
4.   Initialize the iteration counter i = 0.
5.   Initialize the current segmented region Pi with all non-zero voxels.
6.   Compute the initial median intensity of Pi.
7.   Set a small convergence threshold epsilon (e.g., epsilon = 0.01).
8.   While true:
9.     i = i + 1
10.    Compute a dynamic threshold based on order statistics of the current region P(i-1).
11.    Create a new region Pi by removing voxels from P(i-1) with intensities below the threshold.
12.    If (the median of Pi (Mdi) is within a small change from its mean (Mi); (|Mdi - Mi| < epsilon):
13.      Output Pi as the final segmented tumor region.
14.      Break the loop.
15.    End If
16.  End While
17. End For
```

4. Results

The effectiveness of a tumor segmentation algorithm is measured using four parameters: true positive (TP), true negative (TN), false positive (FP), and false negative (FN) (as illustrated in Fig. 4). TP and TN represent the number of correctly identified tumor and non-tumor pixels, respectively. FN and FP indicate the number of wrongly classified tumor and non-tumor pixels. AP and AN are the total number of actual tumor and non-tumor pixels, while PP and PN are the total number of predicted tumor and non-tumor pixels.

Normalization of the parameter associated with the main class on the main diagonal can be achieved through two methods: row normalization and column normalization. Row normalization of this element indicates how sensitive the method is to the segmentation of all tumor voxels, making it a crucial criterion for evaluating the efficiency of the algorithm directly. As shown in Formula 1, this index has a direct one-to-one relationship with TP. Since TP + FN equals AP, a higher SN leads to an

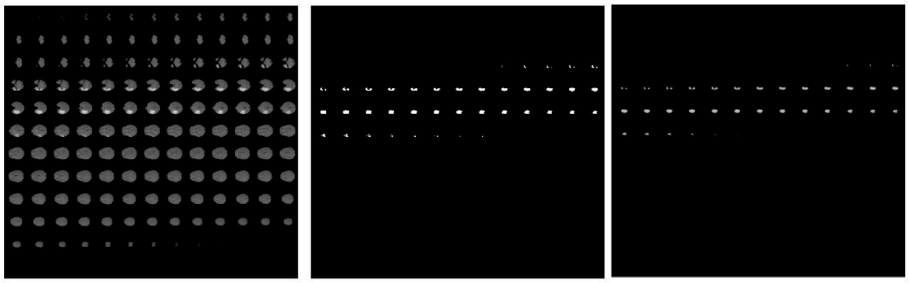


Fig. 3. Visual comparison of the tumor regions extracted by the proposed adaptive method and their related reference regions.

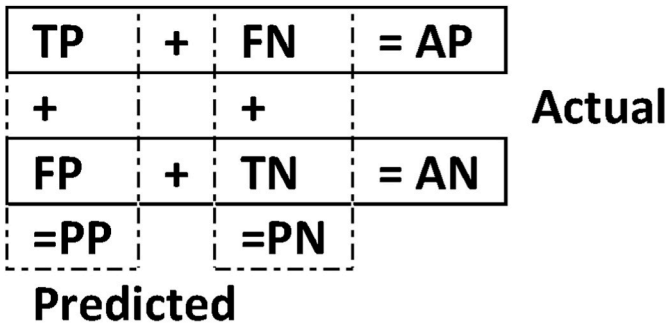


Fig. 4. Complete description of the tumor segmentation algorithm performance.

increase in TP while simultaneously decreasing FN. This results in better diagonalization of the row associated with the main class. The importance of this index makes it a valuable measure for assessing the sensitivity of the tumor detection algorithm and reflects the clinical diagnostic value of the method's results. The column normalization of this element is also a symbol of the positive predictive value of the algorithm in tumor detection.

$$\begin{cases} SN = \dfrac{TP}{AP} = \dfrac{TP}{TP + FN} \\ PPV = \dfrac{TP}{PP} = \dfrac{TP}{TP + FP} \end{cases} \tag{6}$$

Normalization of the parameter related to the complementary class on the main diameter can be achieved in two ways: row normalization and column normalization. Row normalization serves as an indicator of the degree of specificity of the method when segmenting all non-tumor voxels and is the most important criterion for indirectly or complementarily assessing the efficiency of the algorithm. On the other hand, column normalization indicates the negative predictive value of the algorithm in tumor detection. When using these two criteria, it should be noted that the large volume of non-tumor areas causes non-boundary values (not close to one) of these two complementary criteria to be of little value for evaluating the tumor segmentation algorithm. In other words, a small tumor volume will essentially cause the relative TN to be higher than FP or FN in any method presented for tumor segmentation, and therefore, these two indicators will be high in any tumor detection algorithm. As illustrated in Formula 7, there is a direct one-to-one relationship between this index and TN. Given the relationship TN + FP = AN, a higher SP results in a higher TN and a lower FP, demonstrating the diagonalization of the row corresponding to the complementary class. This characteristic makes this index the most valuable for evaluating the specificity of the tumor detection algorithm and, ultimately, the effectiveness of the method in terms of clinical treatment.

$$\begin{cases} SP = \dfrac{TN}{AN} = \dfrac{TN}{TN + FP} \\ NPV = \dfrac{TN}{PN} = \dfrac{TN}{TN + FN} \end{cases} \tag{7}$$

So far, we have identified four indicators to assess the performance of a tumor detection algorithm based on four evaluation parameters. In the context of a two-class classification, it may be beneficial to use a combined indicator. We have three combined indicators for this purpose: accuracy, the Matthews correlation coefficient (MCC), which is calculated using all four parameters shown in Fig. 1; and the F1-measure index, calculated using three of these parameters.

All three of these indicators can provide reliable comparisons for tumor detection algorithms when their values are close to one. A high value for these indicators indicates a low false positive (FP) rate and a high true positive (TP) rate, highlighting the clinical value of the tumor detection algorithm. However, at non-marginal values (i.e., values below 98 %), these combined indicators tend to reflect diagnostic accuracy more than reliability in tumor detection. This means that while an increase in these indicators can lead to a decrease in FP, it does not guarantee an increase in TP.

This issue arises from the nature of tumor segmentation, where the non-tumor area is approximately a hundred times larger than the tumor area. Consequently, the effects of TN and FP on the combined indices are more pronounced due to their larger values compared to TP and FN. As a result, in their non-marginal states, combined indices for evaluating tumor detection tend to prioritize reducing FP rather than increasing TP, which does not fully align with the challenges of tumor detection. The HD95 index faces similar limitations in its non-marginal values, where decreases do not reliably indicate improved tumor detection or an increase in TP.

$$F1 - measure = 2 \times \frac{PPV \times SN}{PPV + SN} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{8}$$

$$\begin{cases} ACC = \dfrac{TP + TN}{AP + AN} = \dfrac{TP + TN}{PP + PN} \\ MCC = \dfrac{TP \times TN - FP \times FN}{\sqrt{AP \times AN \times PP \times PN}} \end{cases} \tag{9}$$

Table 2 displays various methods for brain tumor segmentation, including algorithm-based and artificial intelligence-based approaches, as well as the datasets used and their reported results. Sensitivity (SN) has been utilized more frequently as a direct evaluation criterion, while specificity (SP) has been used more often as an indirect evaluation criterion. The accuracy (ACC) criterion has been the most commonly used among the combined evaluation criteria. It is important to note that these indicators are reported as single values for the entire dataset. These values provide an overall performance assessment for the dataset but do not guarantee the algorithm's performance for each individual within the dataset.

In Table 2, the combination of support vector machine and fuzzy inference [43] reported a slightly higher accuracy than the proposed

Table 2
Methods for brain tumor segmentation, including algorithm-based and artificial intelligence-based approaches, as well as the datasets used and their reported results.

[Ref] Method (year)	BraTS 2019/2020			BraTS 2023/2024		
	ACC	SN	SP	ACC	SN	SP
[43] SVM and Fuzzy Classifier (2022)	99.2					
[51] A hybrid approach using UNet (2024)						
[52] GAIR-U-Net (2024)		95.61	98.97			
[53] DSUNET (2024)	98.67	98.56	98.77			
[54] BTS U-Net (2025)	93.2	97.3	77.6			
[55] Learning multi-modal (2023)	87.31					
[56] BTSegDiff (2025)		93.66	99.47			
[57] stacking ensemble learning (2025)	98.98	**98.73**	98.9	98.1	**98.2**	98.3
[58] SFM optimizer with U-net (2025)		92.49	98.81			
[59] Advanced CNN Architecture (2025)				98.47	98.55	**99.52**
[60] Federated Learning (2023)		90	95			
[61] RU-Net2+ (2023)	98.51		97.16			
Our Method	**99.1**	98	**99.1**	**99.07**	96.81	99.08

method on the same data set, despite only using the ACC combined index and not presenting the SN and SP indices. However, all deep learning methods on the same data set showed less efficiency than the proposed method. Based on these details, the proposed method is relatively comparable to other methods, especially those based on deep learning. Despite its simple algorithm and calculations based only on intensity values, the proposed method enhances the spatial correlation of the tumor in the pre-processing stage, facilitating tumor segmentation connectivity.

The indicators mentioned in the previous paragraphs are used for evaluation. Additionally, the values of these indicators were calculated for the entire dataset as well as for each person in the dataset. Subsequently, the histogram of these values was drawn to ensure the proper performance of the tumor segmentation algorithm on each individual in the dataset.

Histograms of the main indicator for combined assessing the quality of brain tumor segmentation (ACC) are shown for 76 LGG and 259 HGG patients, and GLI patients in BraTS20023 in Fig. 5. These experimental results show that our analytical method achieves stable segmentation performance without any outliers. Since the evaluation is performed on the entire dataset rather than a subset, the results offer higher reliability and robustness, which can be valuable for potential clinical applications.

In Fig. 5, it is evident that the proposed method achieves a minimum accuracy of over 97 % across all 1585 patients in the dataset. It's important to note that when evaluating the generalizability of machine learning methods, only a small portion of the dataset is used for testing. The majority of data diversity is utilized for training, resulting in less diversity in the test set. Therefore, the performance displayed in Fig. 5 is highly promising for an adaptive method that leverages physician knowledge.

While accuracy is an important indicator that often yields high values, comparing accuracy relative to other methods can serve as an effective criterion for assessing their performance. Fig. 5 illustrates that the proposed method attained an accuracy of slightly less than 98 % on only 61 out of 1585 images. This means that for 96.2 % of the images, the proposed method delivered an accuracy that was greater than or equal to the average accuracy reported for recently developed deep networks in Table 2, for only about 10 % (test data) of these images. It is also important to note that the proposed method analyzes the statistical characteristics of each 3D image individually and does not rely on any probabilistic models, which may introduce uncertainty. In contrast,

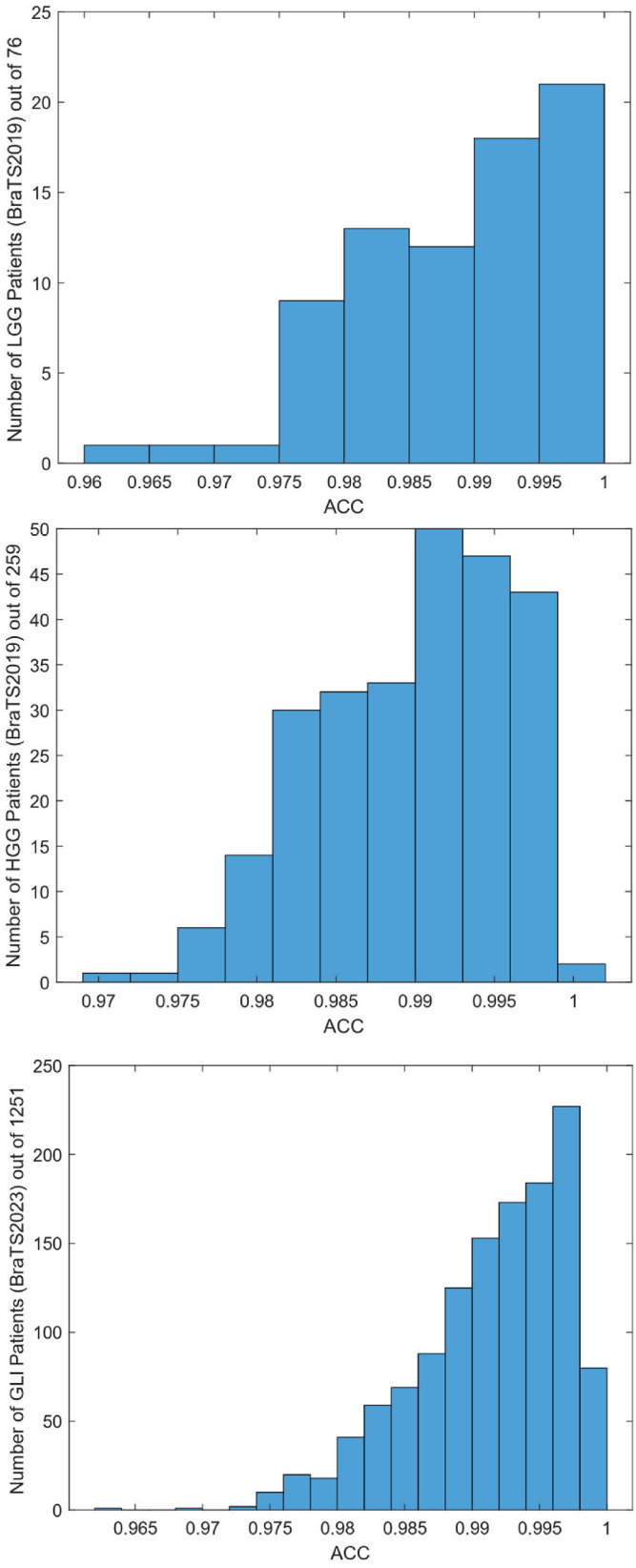


Fig. 5. Histogram of accuracy (ACC) for assessing segmentation performance across 76 LGG and 259 HGG patients from BraTS2019 and 1251 GLI patients from BraTS2023.

results from deep networks deal with uncertainty, leading to variability in performance metrics from one execution to another.

5. Discussion

This study proposed an adaptive brain tumor segmentation algorithm inspired by expert visual perception. Despite its algorithmic simplicity and lack of training requirements, the method delivered consistently high performance on both the BraTS2019 and BraTS2023 datasets, with accuracy, sensitivity (SN), and specificity (SP) metrics confirming its effectiveness.

Notably, in less than 5 % of the dataset (around 60 out of 1251 cases), the segmentation accuracy fell slightly below 98 %. These outlier cases may be attributed to instances of false positives or tumor regions with non-uniform intensity profiles. While the performance in these instances is still comparable to the average reported accuracy of recent deep learning methods, we recognize the need for deeper visual analysis and tailored preprocessing or postprocessing strategies to mitigate such edge-case errors.

In deep learning approaches, the dataset is typically divided into training (70 %), testing (20 %), and validation (10 %) subsets, and the reported results are limited to a portion of the dataset. Our analytical method requires no training phase, and the evaluation is conducted directly on the entire dataset, ensuring that the results are naturally generalizable and valid. Therefore, the performance of our approach is not only competitive with, but in some cases superior to, deep learning models.

6. Limitations of the study

While the proposed method offers several advantages in terms of interpretability, computational simplicity, and independence from large training datasets, there are important limitations to consider:

- Intensity Dependency: The algorithm relies on intensity-based heuristics and may underperform in cases where tumor and non-tumor tissues exhibit similar intensity values, especially in noisy or low-contrast scans.
- Lack of Private Clinical Data: Evaluation was limited to publicly available datasets (BraTS2019 and BraTS2023). Variability in acquisition protocols, scanner types, and patient populations in clinical environments may impact segmentation performance.

Real-World Dataset Validation.

We acknowledge that real-world clinical MRI data may include diverse imaging artifacts and patient-specific features not present in BraTS datasets. However, due to the unavailability of access to private hospital imaging repositories, validation on non-public clinical datasets was not feasible at this stage. Future work will prioritize collaborations with clinical institutions to address this gap and ensure generalizability in practical deployment settings.

7. Computational complexity and runtime comparison

A major advantage of our method lies in its suitability for real-time applications and resource-constrained platforms. Unlike deep learning models that require high-end hardware for training and inference, our technique operates entirely in a one-dimensional histogram domain. It avoids any learning phase, GPU acceleration, and consumes significantly more memory.

From a computational standpoint, our method exhibits a time complexity of O(N log N) per volume (where N is the number of voxels), primarily due to histogram sorting and pruning. In contrast, CNN-based models involve $O(N \times k^2)$ operations across multiple convolutional layers (where k represents the kernel size used in convolution layers), increasing the processing burden, especially for 3D data. These comparisons support the claim that the proposed algorithm is lightweight, fast, and suitable for edge computing environments and embedded diagnostic tools.

Although the algorithm is lightweight and training-free compared to deep learning models, it still requires non-negligible processing time for full 3D volumes.

8. Comparative evaluation with deep learning models

Many high-performing deep learning methods achieve impressive accuracy through extensive training on large labeled datasets. However, they are susceptible to overfitting and may lack generalizability across unseen domains. Our method, by contrast, delivers stable, training-free performance on full benchmark datasets.

Fig. 3 demonstrates the effectiveness of our method in visually identifying slides containing tumor regions, providing a visual validation of segmentation quality. Table 2 reports a detailed comparison between our proposed method and recent state-of-the-art (SoTA) approaches in terms of correctly identifying tumor pixels as a quantitative benchmarking. While the SoTA methods achieve higher performance on their reported subsets, it should be noted that these results are typically based on only ~20 % of the dataset (held-out test sets). As shown in Fig. 5, it consistently delivers stable results without a single outlier, demonstrating robustness and generalizability that are not always evident from limited subset evaluations.

Recent studies (e.g., Santhosh et al., 2025; Kumar et al., 2025) reported validation accuracies of 98.6 % and 98.9 %, respectively, using deep architectures such as CNN-BiLSTM and hybrid optimization. Our method achieved a total accuracy of 99.13 % on the full BraTS2023 dataset without training or fine-tuning, underscoring the strength of expert-inspired preprocessing and deterministic segmentation.

Benchmarking against SoTA underlines competitiveness, while full-dataset evaluation highlights the robustness and reliability of our approach.

In addition, while deep models often require retraining or adaptation when applied to datasets with varying modalities, our method remained robust across both BraTS2019 and BraTS2023 datasets without any algorithmic modification, further reinforcing its potential for scalable deployment.

9. Implications

The results suggest that perceptually inspired, deterministic segmentation techniques can be credible alternatives to complex deep learning models in certain diagnostic workflows. Their simplicity, transparency, and speed make them particularly well-suited for integration into real-time systems, low-power devices, and resource-constrained healthcare environments. Such methods also promote interpretability, which is essential for clinical acceptance and trust.

10. Conclusions

This work presented a novel, adaptive pruning-based approach to brain tumor segmentation in 3D MRI images. Without relying on training or large datasets, the method delivers competitive segmentation performance when compared to deep learning models, while offering superior computational efficiency and interpretability.

The proposed algorithm demonstrates that intelligent, perceptually guided segmentation can support the development of scalable, patient-centric, and accessible medical technologies, particularly in environments where speed, transparency, and resource limitations are critical.

CRediT authorship contribution statement

Ali Mehrabi: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Nasser Mehrshad:** Validation,

Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Ethical statement for solid state ionics

1) This material is the authors' own original work, which has not been previously published elsewhere.
2) The paper is not currently being considered for publication elsewhere.
3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
5) The results are appropriately placed in the context of prior and existing research.
6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving a proper reference.
7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.

Author declaration on AI use

The authors declare that no generative AI or AI-assisted technologies were used in the writing or editing of this manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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